Lab Sheet 1: DDL Commands – CREATE, ALTER and DROP

SQL Answers and Expected Outputs (MySQL Workbench)

Important: The outputs below are typical MySQL Workbench messages. Exact row counts/messages may vary by MySQL version and existing database state. Some tasks ask for alternative table definitions; do not run duplicate CREATE TABLE statements for a table that already exists. Use the relevant definition or a fresh database.

1. Create a database named CollegeDB.

SQL Query:

```
CREATE DATABASE CollegeDB;
USE CollegeDB;
```

Expected Output:

```
Query OK, 1 row affected
Database changed
```

2. Create a Student table with columns: Student_ID, Name, Age, Course, and City.

SQL Query:

```
CREATE TABLE Student (
  Student_ID INT,
  Name VARCHAR(100),
  Age INT,
  Course VARCHAR(50),
  City VARCHAR(50)
);
```

Expected Output:

```
Query OK, 0 rows affected
```

3. Create an Employee table with Emp_ID, Emp_Name, Department, and Salary.

SQL Query:

```
CREATE TABLE Employee (
  Emp_ID INT,
  Emp_Name VARCHAR(100),
  Department VARCHAR(50),
  Salary DECIMAL(10,2)
);
```

Expected Output:

```
Query OK, 0 rows affected
```

4. Create a Department table with Dept_ID, Dept_Name, and Location.

SQL Query:

```
CREATE TABLE Department (
  Dept_ID INT,
  Dept_Name VARCHAR(100),
  Location VARCHAR(100)
);
```

Expected Output:

```
Query OK, 0 rows affected
```

5. Create a Product table with Product_ID, Product_Name, Price, and Quantity.

SQL Query:

```
CREATE TABLE Product (
  Product_ID INT,
  Product_Name VARCHAR(100),
  Price DECIMAL(10,2),
  Quantity INT
);
```

Expected Output:

```
Query OK, 0 rows affected
```

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6. Create a Customer table with Customer_ID, Customer_Name, Email, and Phone.

SQL Query:

```
CREATE TABLE Customer (
  Customer_ID INT,
  Customer_Name VARCHAR(100),
  Email VARCHAR(150),
  Phone VARCHAR(20)
);
```

Expected Output:

```
Query OK, 0 rows affected
```

7. Create a Course table with Course_ID, Course_Name, Duration, and Fees.

SQL Query:

```
CREATE TABLE Course (
  Course_ID INT,
  Course_Name VARCHAR(100),
  Duration VARCHAR(50),
  Fees DECIMAL(10,2)
);
```

Expected Output:

```
Query OK, 0 rows affected
```

8. Create a Faculty table with Faculty_ID, Faculty_Name, Subject, and Salary.

SQL Query:

```
CREATE TABLE Faculty (
  Faculty_ID INT,
  Faculty_Name VARCHAR(100),
  Subject VARCHAR(100),
  Salary DECIMAL(10,2)
);
```

Expected Output:

```
Query OK, 0 rows affected
```

9. Create a Library table with Book_ID, Book_Name, Author, and Price.

SQL Query:

```
CREATE TABLE Library (
  Book_ID INT,
  Book_Name VARCHAR(150),
  Author VARCHAR(100),
  Price DECIMAL(10,2)
);
```

Expected Output:

```
Query OK, 0 rows affected
```

10. Create a Department table with Dept_ID as the Primary Key.

SQL Query:

```
-- Use this definition instead of the earlier Department table
CREATE TABLE Department (
  Dept_ID INT PRIMARY KEY,
  Dept_Name VARCHAR(100),
  Location VARCHAR(100)
);
```

Expected Output:

```
Query OK, 0 rows affected
Note: If Department already exists, use ALTER TABLE Department ADD PRIMARY KEY (Dept_ID) after ensuring Dept_ID has no
```

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11. Create a Student table with Student_ID as the Primary Key and Name as NOT NULL.

SQL Query:

```
CREATE TABLE Student (
  Student_ID INT PRIMARY KEY,
  Name VARCHAR(100) NOT NULL,
  Age INT,
  Course VARCHAR(50),
  City VARCHAR(50)
);
```

Expected Output:

```
Query OK, 0 rows affected
Note: If Student already exists, use a new table name or alter the existing table instead.
```

12. Create an Employee table with a UNIQUE constraint on Email.

SQL Query:

```
CREATE TABLE Employee (
  Emp_ID INT PRIMARY KEY,
  Emp_Name VARCHAR(100),
  Department VARCHAR(50),
  Salary DECIMAL(10,2),
  Email VARCHAR(150) UNIQUE
);
```

Expected Output:

```
Query OK, 0 rows affected
Note: This is an alternative Employee definition that includes Email.
```

13. Create a Product table with a CHECK constraint that Price > 0.

SQL Query:

```
CREATE TABLE Product (
  Product_ID INT PRIMARY KEY,
  Product_Name VARCHAR(100),
  Price DECIMAL(10,2) CHECK (Price > 0),
  Quantity INT
);
```

Expected Output:

```
Query OK, 0 rows affected
Note: This is an alternative Product definition. MySQL 8.0.16+ enforces CHECK constraints.
```

14. Add a new column Email to the Student table using ALTER TABLE.

SQL Query:

```
ALTER TABLE Student ADD COLUMN Email VARCHAR(150);
```

Expected Output:

```
Query OK, 0 rows affected
```

15. Add a Phone column to the Student table.

SQL Query:

```
ALTER TABLE Student ADD COLUMN Phone VARCHAR(20);
```

Expected Output:

```
Query OK, 0 rows affected
```

16. Modify the datatype/size of the Name column in the Student table.

SQL Query:

```
ALTER TABLE Student MODIFY COLUMN Name VARCHAR(150);
```

Expected Output:

```
Query OK, 0 rows affected
```

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17. Rename the Student table to Student_Details.

SQL Query:

```
RENAME TABLE Student TO Student_Details;
```

Expected Output:

```
Query OK, 0 rows affected
```

18. Rename the column City to Address.

SQL Query:

```
ALTER TABLE Student_Details RENAME COLUMN City TO Address;
```

Expected Output:

```
Query OK, 0 rows affected
MySQL 8.0+ syntax.
```

19. Add a Salary column to the Employee table.

SQL Query:

```
ALTER TABLE Employee ADD COLUMN Salary DECIMAL(10,2);
```

Expected Output:

```
Query OK, 0 rows affected
If Salary already exists in the table, skip this operation.
```

20. Drop the Phone column from the Student table.

SQL Query:

```
ALTER TABLE Student_Details DROP COLUMN Phone;
```

Expected Output:

```
Query OK, 0 rows affected
```

21. Drop the Email column from the Student table.

SQL Query:

```
ALTER TABLE Student_Details DROP COLUMN Email;
```

Expected Output:

```
Query OK, 0 rows affected
```

22. Remove all records from a table using TRUNCATE.

SQL Query:

```
TRUNCATE TABLE Student_Details;
```

Expected Output:

```
Query OK, 0 rows affected
The table structure remains, but all rows are removed.
```

23. Drop the Course table from the database.

SQL Query:

```
DROP TABLE Course;
```

Expected Output:

```
Query OK, 0 rows affected
```

24. Drop the Customer table from the database.

SQL Query:

```
DROP TABLE Customer;
```

Expected Output:

```
Query OK, 0 rows affected
```

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25. Create a complete Student table using Primary Key, NOT NULL, UNIQUE and CHECK constraints, and demonstrate suitable ALTER and DROP operations.

SQL Query:

```
CREATE TABLE Student_Complete (
  Student_ID INT PRIMARY KEY,
  Name VARCHAR(100) NOT NULL,
  Email VARCHAR(150) UNIQUE,
  Age INT CHECK (Age >= 16),
  Course VARCHAR(50),
  City VARCHAR(50)
);

ALTER TABLE Student_Complete
  ADD COLUMN Phone VARCHAR(20);

ALTER TABLE Student_Complete
  DROP COLUMN Phone;
```

Expected Output:

```
Query OK, 0 rows affected for each statement
The table is created with constraints, Phone is added, and then Phone is removed.
```

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